REVISION WORKSHEET

Exam Skills Drills

Quick reference for the four topics students most often drop marks on. Practise each interactively in the online lab.

Subnetting & VLSM

Usable hosts = $2^{(32 - \text{prefix})} - 2$ (lose the network ID and broadcast). The network address falls on a multiple of the block size.

Prefix	Mask	Block	Usable hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

VLSM: give each subnet the smallest block that fits its host need; allocate the largest requirement first.

Ports & protocols

Protocol	Port	Use
HTTP	80	web (unencrypted)
HTTPS	443	secure web
FTP	21	file transfer
SSH	22	secure remote login
Telnet	23	remote login (unencrypted)
SMTP	25	sending email
DNS	53	name → IP
DHCP	67/68	automatic IP config
POP3	110	retrieving email
RDP	3389	remote desktop

TCP = connection-oriented, reliable. **UDP** = connectionless, faster. Both work at the Transport layer (OSI 4).

Troubleshooting from command output

Command	Tells you
`ipconfig`	IP, mask, gateway. 169.254.x.x (APIPA) = DHCP failed.
`ping`	Reachability. "Request timed out" vs "Destination host unreachable" (no route).

`tracert`	Where along the path it stops.
`nslookup`	Whether DNS resolves a name.
`show ip interface brief`	Interface IPs and up/down status.

Method: work along the path (host → gateway → next hop → destination) until a test fails — the fault is there.

ACLs

An ACL is an **ordered** list of permit/deny rules, checked **top to bottom**; the router acts on the **first match**. Anything unmatched hits the **implicit deny** at the end. Order matters — a broad rule above a specific one can stop the specific rule ever being used.